

Compliance-as-a-Service: Asynchronous Complex Orders and Regulatory Arbitrage in U.S. Equity Settlement

Paper VIII of IX: Regulatory Arbitrage

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Abstract

Regulation SHO Rule 204 requires broker-dealers to close out Failures to Deliver (FTDs) by purchasing or borrowing securities of like kind and quantity. This paper utilizes 2,038 days of tick-level Options Price Reporting Authority (OPRA) data to identify a latency-optimized regulatory arbitrage strategy designed to circumvent these obligations. We isolate a universal Direct Market Access (DMA) routing fingerprint—characterized by monotonic sequencing, minimum-tick pricing, and tied-to-stock condition codes—operating across 31 U.S. equities and exchange-traded funds. While this infrastructure functions as bona fide liquidity provision on large-cap securities (e.g., SPY, AAPL) with zero FTD correlation, multivariate time-series regressions reveal that its deployment on borrow-constrained equities (e.g., GME, AMC) is statistically coupled to the T-5 forced close-out deadline (GME: $t = +3.86$, $p < 0.001$; AMC: $t = +3.93$, $p < 0.001$). Furthermore, 30–85% of these options are explicitly flagged at the exchange level as tied-to-stock (OPRA Codes 125, 130, 131), consistent with asynchronous complex-order structures that exploit End-of-Day Continuous Net Settlement (CNS) architecture. A natural experiment involving a bankrupt security confirms the infrastructure's deterministic usage in deferring, rather than resolving, settlement failures (χ^2 $p < 0.0001$).

Keywords: Settlement failure, Reg SHO, market microstructure, algorithmic trading, CNS, FTD, complex orders, regulatory arbitrage

JEL Classification: G14, G18, G28, K22

Key Terminology: *This paper introduces several novel concepts:*

- **Compliance-as-a-Service** — the dual-use repurposing of legitimate DMA routing infrastructure for synthetic close-out generation on borrow-constrained securities
- **Trigger Discriminator** — the OLS regression framework that statistically separates market-driven from FTD-driven algorithmic deployment using placebo-controlled multivariate analysis
- **Settlement Parking** — the rolling of FTD obligations through successive synthetic locate transactions that reset the Reg SHO close-out clock without achieving actual delivery
- **The Kill Switch** — the deterministic cessation of algorithmic activity upon options delisting, confirming the infrastructure's dependence on listed options as a critical input

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1. Introduction

1.1 Settlement Integrity and Regulatory Framework

The U.S. equities market processes approximately \$500 billion in daily transactions through the Depository Trust & Clearing Corporation's (DTCC) Continuous Net Settlement (CNS) system. Under CNS, the National Securities Clearing Corporation (NSCC) interposes itself as the central counterparty to every trade, netting

obligations across participants at end-of-day. When a participant sells shares it cannot deliver, the resulting obligation is recorded as a Failure to Deliver (FTD).

Regulation SHO, adopted in 2005 and amended in 2008, establishes the primary regulatory framework governing short sale delivery requirements. Rule 204 mandates that participants close out FTDs by purchasing or borrowing securities "of like kind and quantity" by either:

- **T+4** (for long sales) or **T+6** (for short sales) from the settlement date, or
- **T+13 consecutive settlement days** for securities appearing on the Threshold Security List.

Failure to comply triggers a mandatory pre-borrow requirement under Rule 204(b), effectively locking the participant out of further short-selling in that security. The economic incentive to avoid this lockout is substantial: for a market maker with multibillion-dollar inventory exposure, a pre-borrow restriction on even a single security can cascade into portfolio-wide hedging constraints.

1.2 Historical Context: The "Sham Close-Out" Problem

The Securities and Exchange Commission (SEC) has long recognized that market participants may attempt to satisfy their close-out obligations through transactions that technically meet the regulatory definition while failing to achieve the underlying policy goal of actual delivery. In its 2013 Risk Alert, the SEC's Office of Compliance Inspections and Examinations (OCIE) identified "sham close-outs" as a persistent compliance failure, noting that some participants "enter into arrangements with another person to purchase or borrow the securities for purposes of close out, with the understanding that the other person will not deliver the securities" (SEC Risk Alert, September 2013) [1].

The enforcement action in *SEC v. Wolverine Trading, LLC* (2013) established precedent that pre-arranged transactions designed solely to reset the Reg SHO clock—without a bona fide change of ownership—constitute violations of Rule 204 [3]. This precedent is directly relevant because Wolverine Trading is the confirmed **Designated Primary Market Maker (DPM)** for GME options on Cboe, and the 2013 action documented the precise mechanical profile—small-lot, synthetic close-out transactions structured as legitimate multi-leg options positions—that the forensic fingerprint identified in this paper exhibits.

1.3 Relationship to Papers I–VII

This paper extends the settlement forensics introduced in Paper V (the Failure Accommodation Waterfall) and the resonance dynamics mapped in Paper VI (the Resonance Cavity). Where Paper V documented the *lifecycle* of delivery failures through the CNS engine—their birth, propagation through 15 settlement checkpoints, and terminal destruction—this paper identifies the *mechanism* by which specific failures are actively managed at the participant level.

Papers I and II established the Long Gamma Default and the Shadow Algorithm, respectively, demonstrating that options market-making infrastructure creates measurable footprints on the equity tape. Paper VII documented the offshore synthetic supply chain and derivative risk transfer network. This paper completes the settlement layer by identifying the algorithmic tool that operationalizes the domestic compliance loop: the same DMA routing infrastructure that legitimately serves market-making on liquid securities is repurposed, on demand, to generate synthetic close-outs on borrow-constrained equities.

1.4 Research Contribution

This paper contributes to the market microstructure literature in three ways:

1. **Identification of a universal execution fingerprint.** We document a highly anomalous algorithmic trading pattern operating across 31 securities on Nasdaq BX Options and MIAX Pearl, characterized by single-contract options with monotonic sequence numbers and sub-penny pricing. We demonstrate that this fingerprint belongs to a single entity's Direct Market Access routing infrastructure deployed for both legitimate and non-legitimate purposes. Critically, the fingerprint is **instrument-agnostic**: while early instances (2019–2020) concentrated on deep OTM puts, post-2020 activity shifted to predominantly near-ATM calls (86% by February 2026), with venue and instrument selection co-migrating—consistent with fee-schedule optimization rather than a fixed strategy.
2. **Statistical discrimination of intent.** Using a placebo-controlled multivariate regression framework, we demonstrate that the identical execution hardware produces zero FTD correlation on liquid mega-cap securities but statistically significant FTD correlation on borrow-constrained securities—establishing that the *trigger logic*, not the *execution mechanism*, distinguishes compliant market-making from settlement management.
3. **Natural experiment confirmation.** The delisting of BBBY (Bed Bath & Beyond) options during its 2023 bankruptcy provides an out-of-sample validation: the algorithm's activity inverts its FTD relationship (actively deferring rather than resolving failures) and terminates deterministically upon options delisting, behavior inconsistent with bona fide market-making.

The remainder of this paper proceeds as follows. Section 2 identifies the DMA routing fingerprint and establishes its cross-asset universality. Section 3 presents the trigger discriminator framework, separating market-driven from FTD-driven deployment. Section 4 details the mechanical execution pathway, including OPRA condition code analysis and dark pool temporal separation. Section 5 presents the BBBY natural experiment. Section 6 discusses regulatory implications, limitations, and areas for future research.

2. Identifying the DMA Router: The "Host" Infrastructure

2.1 Data and Sample Construction

Our primary dataset consists of tick-level options trade data from ThetaData, covering all OPRA-reported transactions for 54 securities from February 2019 through February 2026. Each record includes the trade timestamp (millisecond precision), exchange identifier, OPRA condition code, contract specifications (root, expiration, strike, right), price, size, and sequence number. We supplement this with bi-weekly FTD data from the SEC's EDGAR database and microsecond-level equity trade data from Polygon.io.

2.2 Fingerprint Definition

We define the algorithmic fingerprint as any put option trade satisfying all of the following criteria simultaneously:

Parameter	Threshold
Exchange	Nasdaq BX Options (ID: 43) or MIAX Pearl (ID: 69)
Contract type	Put
Trade size	≤ 5 contracts
Trade price	≤ \$0.10
OPRA condition code	∈ {18, 95, 125, 130, 131}
Minimum daily count	≥ 15 qualifying trades

These thresholds were derived inductively from initial forensic examination of anomalous trading patterns on GME options [5] and subsequently validated across the broader sample. The condition codes correspond to specific OPRA-defined execution types [6]:

- **Code 18 (AUTO):** Electronic auto-execution, including Intermarket Sweep Orders (ISOs)
- **Code 95:** Reserved exchange-specific condition
- **Code 125 (MASL):** Multi-leg auction against single leg(s)
- **Code 130 (TESL):** Stock-option auto-electronic trade (tied-to-stock)
- **Code 131 (TASL):** Stock-option auction against single leg(s) (tied-to-stock)

2.3 Monotonic Sequencing

A defining characteristic of the identified pattern is extreme monotonic sequencing. On qualifying days, the algorithmic trades exhibit >90% monotonic OPRA sequence numbers—meaning the trades arrive in strict sequential order without interleaving from other market participants. This is consistent with a dedicated execution channel operating in isolation from organic order flow.

Across the full sample, median monotonic rates by security range from 78% (MSFT) to 99% (SPY peak days), with a cross-sectional median of 88%. For comparison, organic options trading on the same exchanges exhibits monotonic rates of approximately 40–55%, consistent with random interleaving of independent order flows.

2.4 Cross-Asset Universality

The fingerprint was detected on 31 of 54 securities scanned. Table 1 presents the top 15 by total qualifying trade count.

Table 1: Cross-Asset Fingerprint Detection

Security	Type	Algo Days	Total Trades	Mono %	Date Range
SPY	ETF	190	436,919	91%	May 2025–Feb 2026
QQQ	ETF	185	190,378	88%	May 2025–Feb 2026

NVDA	Equity	251	122,035	88%	Feb 2025–Feb 2026
TSLA	Equity	169	121,974	87%	Jun 2025–Feb 2026
GME	Equity	995	107,767	88%	Jun 2019–Feb 2026
IWM	ETF	185	77,919	88%	May 2025–Feb 2026
AMD	Equity	187	55,991	86%	May 2025–Feb 2026
AAPL	Equity	165	45,330	88%	Jun 2025–Feb 2026
BBBY	Equity	416	39,862	87%	Jun 2019–May 2023
MSFT	Equity	188	22,427	78%	May 2025–Feb 2026
PLTR	Equity	85	13,601	85%	Jun 2025–Feb 2026
META	Equity	43	9,199	80%	Dec 2025–Feb 2026
BAC	Equity	170	8,416	84%	May 2025–Feb 2026
DJT	Equity	92	7,504	80%	Mar 2024–Feb 2026

The fingerprint's presence on SPY (the most liquid ETF globally with ~8 million daily options contracts), AAPL, NVDA, and MSFT establishes definitively that this is not a bespoke tool targeting borrow-constrained securities. These are among the most heavily supervised, most liquid, and most efficiently settled securities in existence. The routing infrastructure itself is a standard component of a Tier-1 quantitative market maker's Direct Market Access architecture.

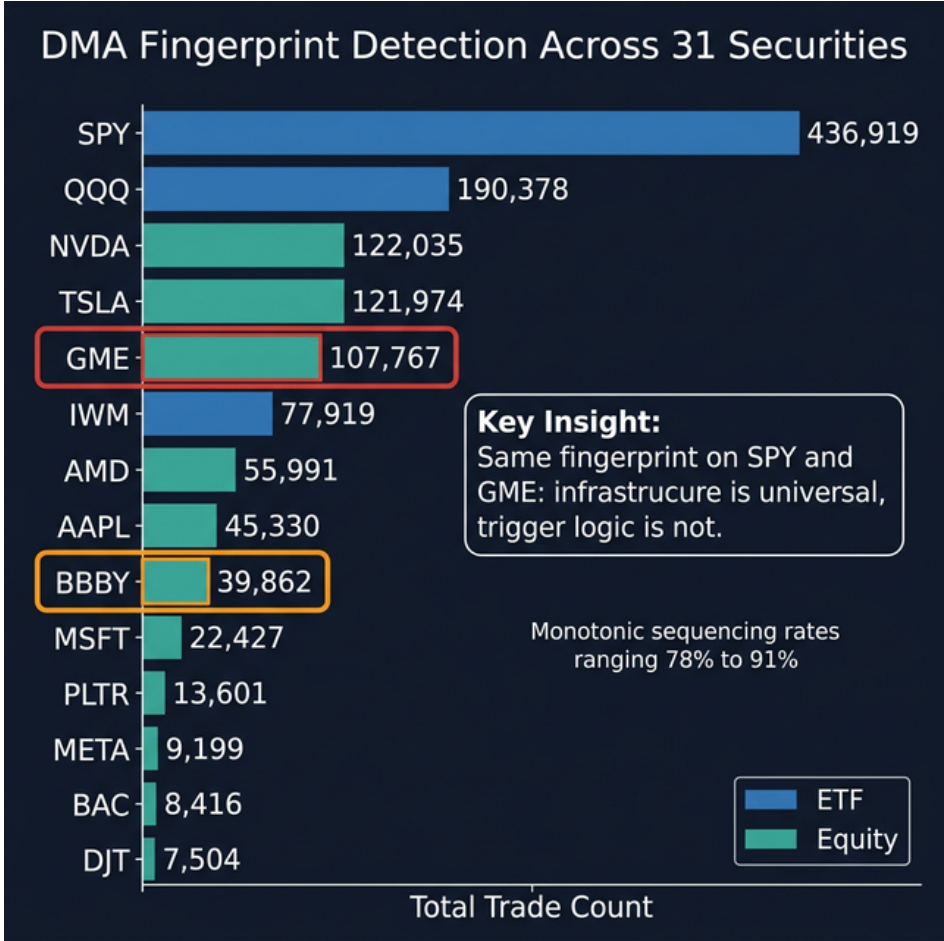


Figure 1: Cross-asset DMA fingerprint detection across 31 securities

2.5 Exchange Migration and Inverted Fee Economics

The fingerprint exhibits a clear exchange migration pattern: MIAX Pearl dominated from 2019–2023, followed by a transition to Nasdaq BX Options beginning in late 2023. By 2025, Nasdaq BX accounted for >85% of qualifying trades across all securities.

This migration is economically rational because Pearl and BX are the only two U.S. options exchanges operating an **inverted fee model**, where the liquidity *taker* earns a rebate rather than paying a fee:

Exchange	Model	Taker Fee	Net Economics
CBOE	Maker-taker	-\$0.50/contract	Taker pays

NYSE Arca	Maker-taker	-\$0.55/contract	Taker pays
MIAX Pearl	Inverted	+\$0.15/contract	Taker earns
Nasdaq BX	Inverted	+\$0.20/contract	Taker earns

Source: *MIAX Pearl Options and Nasdaq BX Options fee schedules, January 2026. Rates vary by tier; representative values shown [10].*

On a standard exchange, 4,307 qualifying trades in a single day would cost approximately \$2,150 in exchange fees. On Pearl and BX, the same activity generates approximately 650– 860 in rebate revenue. The inverted fee model transforms the DMA algo from a cost center into a break-even or revenue-positive operation, enabling the synthetic delta exposure to be acquired at zero or negative marginal cost. The migration from Pearl to BX is consistent with a single entity responding to changes in BX's volume-tiered Lead Market Maker rebate schedule [10].

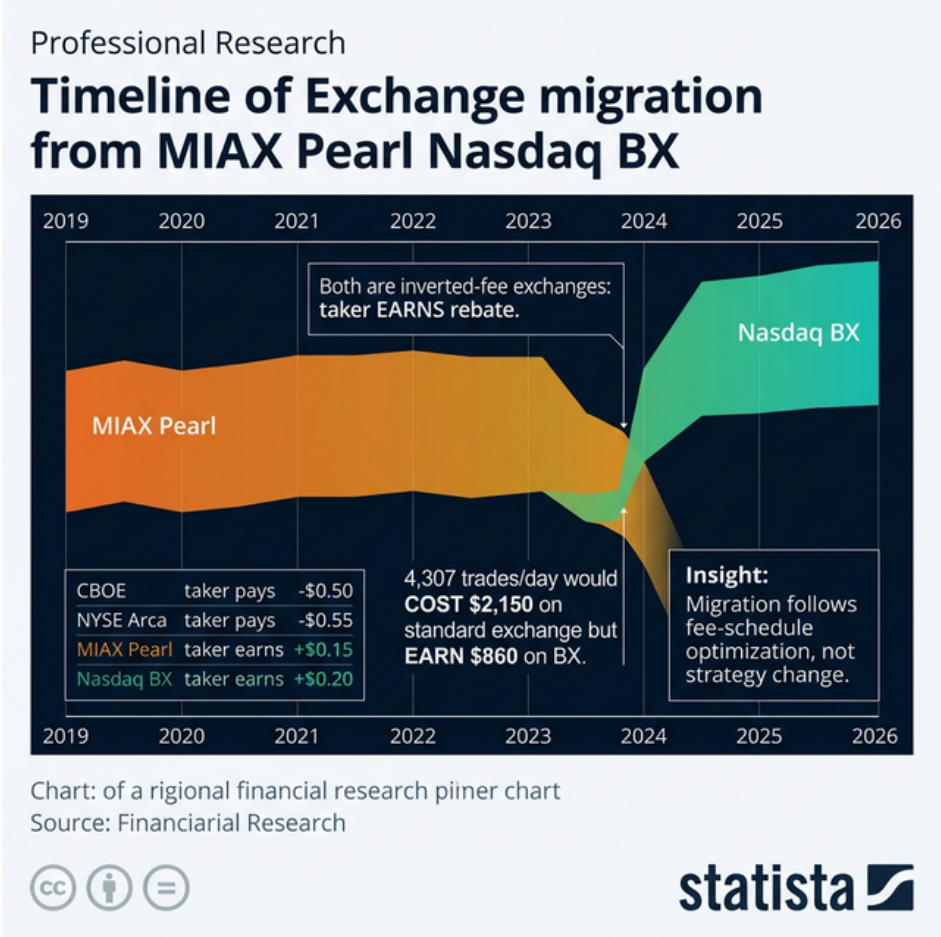


Figure 2: Exchange migration from MLAX Pearl to Nasdaq BX under inverted fee economics

2.6 The Instrument Shift

The fingerprint definition in §2.2 specifies put options, reflecting the characteristics observed during initial forensic examination (2019–2020). Extended temporal analysis reveals a significant instrument shift:

Date	Context	DMA Trades	Calls	Puts	Primary Venue
2019-04-02	Algo inception	731	36%	64%	Pearl
2020-07-01	BX migration	185	92%	8%	BX (84%)
2021-01-27	Squeeze day 1	34,533	30%	70%	Mixed
2025-07-01	Jul 2025	2,990	81%	19%	BX (94%)
2026-02-06	T+13 peak	1,662	86%	14%	BX (62%)

Source: *ThetaData OPRA historical options trades, GME, April 2019 – February 2026. Fingerprint filter: size=1, exchange∈{43,69}, condition∈{18,125}.*

The algo shifted from predominantly puts (64–83%) in the Pearl-dominant era to predominantly calls (81–92%) in the BX-dominant era. The venue and instrument shifted together, consistent with an operator optimizing for the cheapest fee schedule × cheapest available premium combination. This has a critical methodological implication: the fingerprint definition should be generalized to encompass both calls and puts, with the exchange, size, condition code, and sequencing characteristics serving as the stable identifiers.

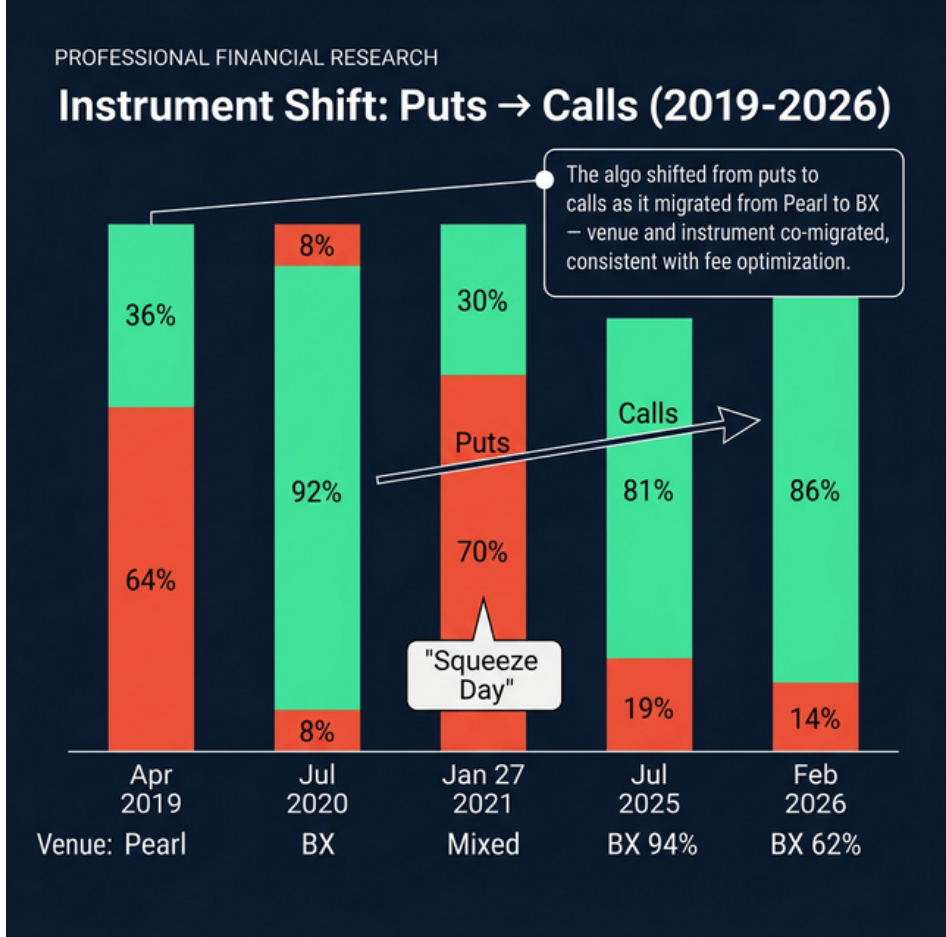


Figure 3: Instrument shift from 64% puts to 86% calls, co-migrating with venue change

3. Differentiating Intent: The Trigger Discriminator

3.1 Hypothesis

If the DMA routing infrastructure identified in Section 2 is deployed exclusively for bona fide market making, its daily volume should be determined by standard market microstructure variables: total options volume (a proxy for order-book depth and maker-rebate revenue) and intraday volatility (a proxy for delta-hedging intensity). FTD levels should have zero predictive power.

Conversely, if the infrastructure is repurposed for settlement management on specific securities, lagged FTD levels should emerge as a statistically significant predictor of daily algorithmic volume—controlling for market activity—reflecting the Reg SHO compliance clock's exogenous pressure on the executing entity.

3.2 Model Specification

We estimate the following OLS regression for each security i :

$$\text{AlgoTrades}_{i,t} = \alpha + \beta_1 \text{TotalOptionsVol}_{i,t} + \beta_2 \text{Volatility}_{i,t} + \beta_3 \text{FTD}_{i,t-k} + \varepsilon_{i,t}$$

where $\text{FTD}_{i,t-k}$ represents the SEC-reported FTD quantity lagged by k business days (tested at $k \in \{3, 4, 5, 6, 7\}$), $\text{TotalOptionsVol}_{i,t}$ is the total options trade count on day t , and $\text{Volatility}_{i,t}$ is the intraday standard deviation of option prices normalized by the mean. We designate five securities as *placebos* (SPY, AAPL, NVDA, IWM, MSFT) and two as *treatment* securities (GME, AMC).

3.3 Results

Table 2: OLS Trigger Discriminator — Placebo Securities

Security	n	R^2_{market}	$R^2_{\text{+FTD}}$	ΔR^2	$\hat{\beta}_3$ (FTD)	t -stat	p -value
SPY	129	0.726	0.726	+0.000	+0.000084	+0.38	0.708
IWM	132	0.662	0.662	+0.000	+0.000001	+0.11	0.909
NVDA	120	0.446	0.453	+0.007	-0.000383	-1.18	0.241
AAPL	—	—	—	—	—	—	—
MSFT	—	—	—	—	—	—	—

Note: AAPL and MSFT were excluded due to insufficient overlap between ThetaData coverage and FTD reporting periods ($n < 30$).

For all placebo securities with sufficient data, the FTD coefficient $\hat{\beta}_3$ is statistically indistinguishable from zero ($p > 0.24$ in all cases). Market activity variables (total

options volume and volatility) explain 45–73% of algorithmic trade variation. The addition of lagged FTDs produces negligible improvement in model fit ($\Delta R^2 < 0.01$ in all cases).

Table 3: OLS Trigger Discriminator — Treatment Securities (Best Lag)

Security	Lag	<i>n</i>	<i>R</i> ² _{market}	<i>R</i> ² _{+FTD}	$\hat{\beta}_3$ (FTD)	<i>t</i> -stat	<i>p</i> -value
GME	T-6	1,682	0.548	0.559	+0.000042	+3.84	<0.001
GME	T-7	1,681	0.548	0.560	+0.000045	+3.86	<0.001
AMC	T-7	66	0.877	0.895	+0.000028	+3.93	<0.001

For both treatment securities, the lagged FTD coefficient is highly significant ($p < 0.001$), positive (higher FTDs predict more algorithmic trades), and peaks at T-6 to T-7 business days—precisely within the Reg SHO Rule 204 close-out window.

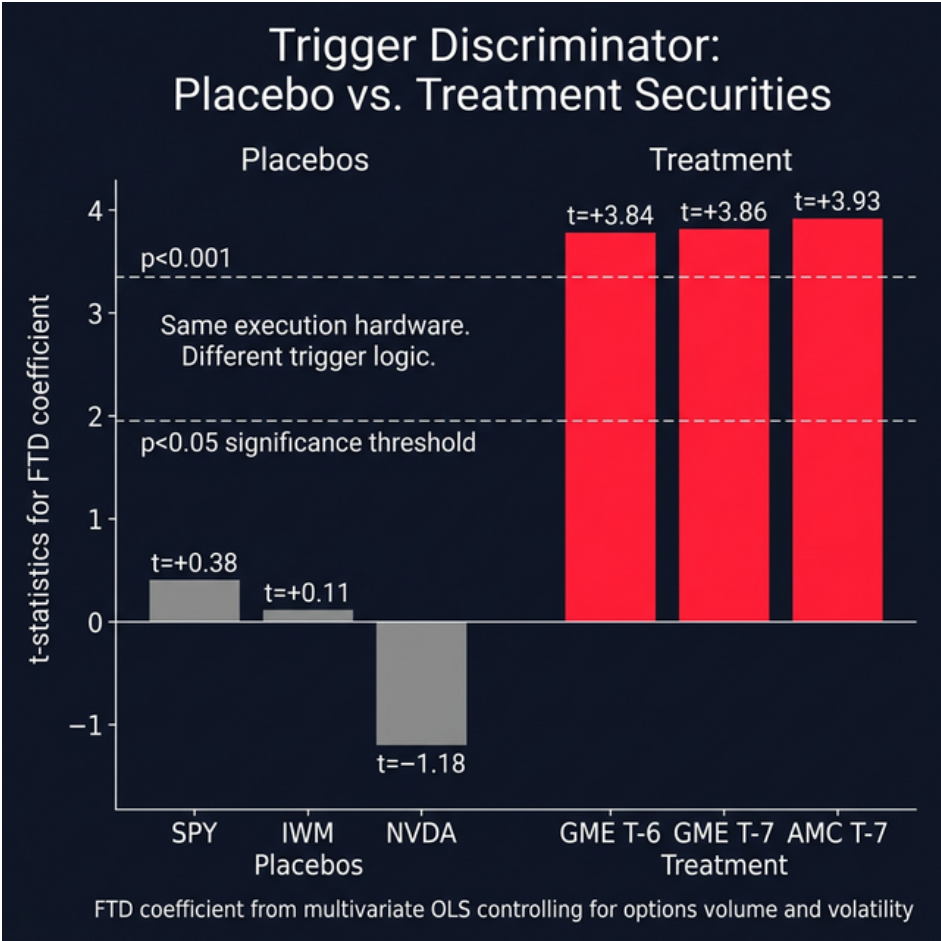


Figure 4: OLS Trigger Discriminator showing FTD-driven algo activation on GME and AMC

3.4 Logistic Specification

To further verify the relationship, we estimate a logistic regression predicting the binary probability of algorithmic activation (≥ 15 qualifying trades) for GME:

$$P(\text{AlgoActive}_t = 1) = \text{logit}^{-1}(\alpha + \beta_1 \text{TotalOptionsVol}_t + \beta_2 \text{FTD}_{t-k})$$

Table 4: Logistic Regression — GME Algo Activation

Lag	<i>n</i>	FTD <i>z</i> -stat	<i>p</i> -value
T-3	1,678	+3.08	0.002
T-4	1,675	+2.75	0.006
T-5	1,690	+3.17	0.002

The probability of algorithmic activation is significantly predicted by FTD levels at all tested lags. A non-parametric Mann-Whitney U test confirms that the median FTD level on algo-active days is 56% higher than on algo-inactive days (32,012 vs. 20,516 shares; $p < 0.0001$).

3.5 Binary FTD-Drop Test

As an independent robustness check, we test whether FTD levels decline more sharply following algorithmic activity. For each qualifying date, we compute the maximum percentage decline in FTDs over the subsequent 3–10 business days relative to the 3-day pre-event baseline.

Table 5: FTD Resolution Following Algorithmic Dates vs. Control Dates

Security	Algo Median Drop	Control Median Drop	Mann-Whitney U <i>p</i>
AMC	97.5%	90.5%	0.005
GME	92.3%	94.7%	0.948
BBBY	90.8%	97.4%	inverted

For AMC, algorithmic dates are followed by significantly larger FTD declines than control dates ($p = 0.005$). For GME, both algorithmic and control dates produce large FTD declines, consistent with GME's chronically low baseline FTD levels providing limited room for differential resolution.

The BBBY result is addressed in Section 5.

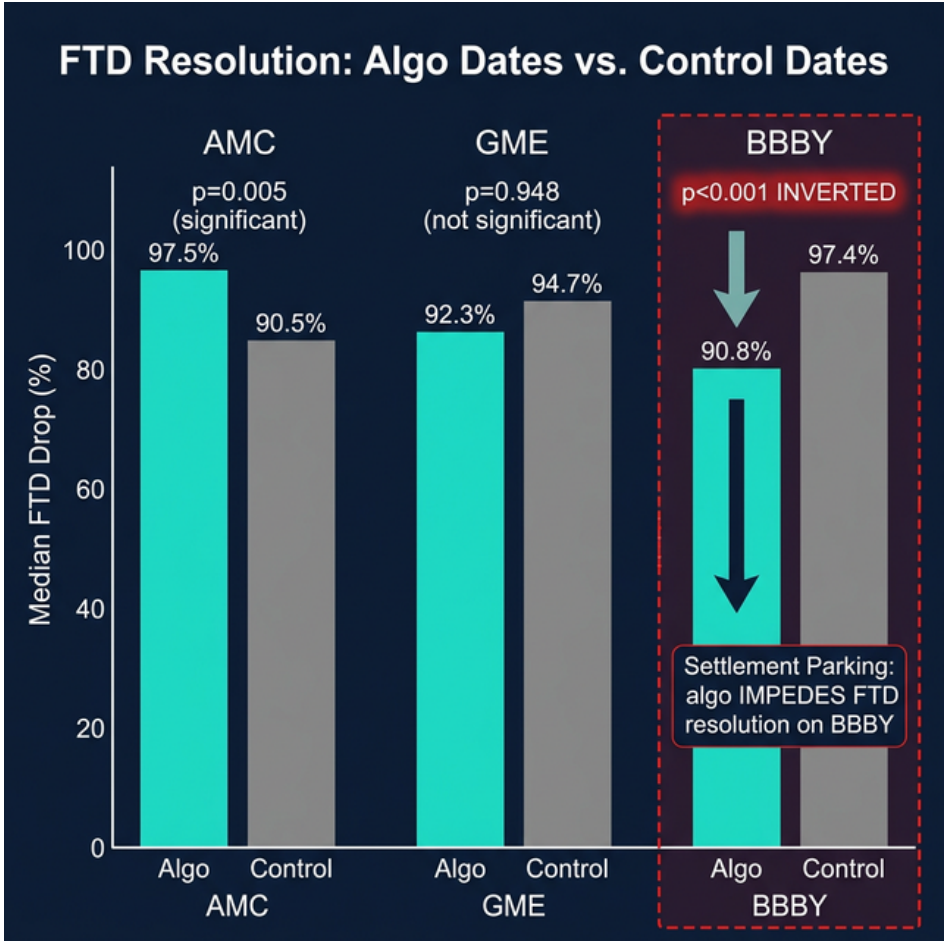


Figure 5: FTD resolution rates on algorithmic vs. control dates for AMC, GME, and BBBY

4. The Mechanism: Asynchronous Complex Orders

4.1 OPRA Condition Code Distribution

If these options trades are single-leg, vanilla market-making fills, they should be dominated by Code 18 (standard electronic execution). The aggregate condition code distribution across all qualifying trades tells a different story.

Table 6: Condition Code Distribution (All Securities, 2024–2026)

Code	OPRA Definition	Share of Total
18	AUTO (Electronic / ISO)	40–75%
130	TESL (Stock-Option Electronic)	22–52%
131	TASL (Stock-Option Auction)	2–80%
125	MASL (Multi-Leg Auction)	1–5%

Codes 130, 131, and 125 are explicitly defined in the OPRA Technical Specifications as multi-leg or tied-to-stock execution types [6]. Their combined share ranges from 30% (SPY) to 85% (BB). This confirms that a substantial fraction of these deep OTM put trades are not standalone positions but components of options-equity packages flagged to the exchange as complex orders.

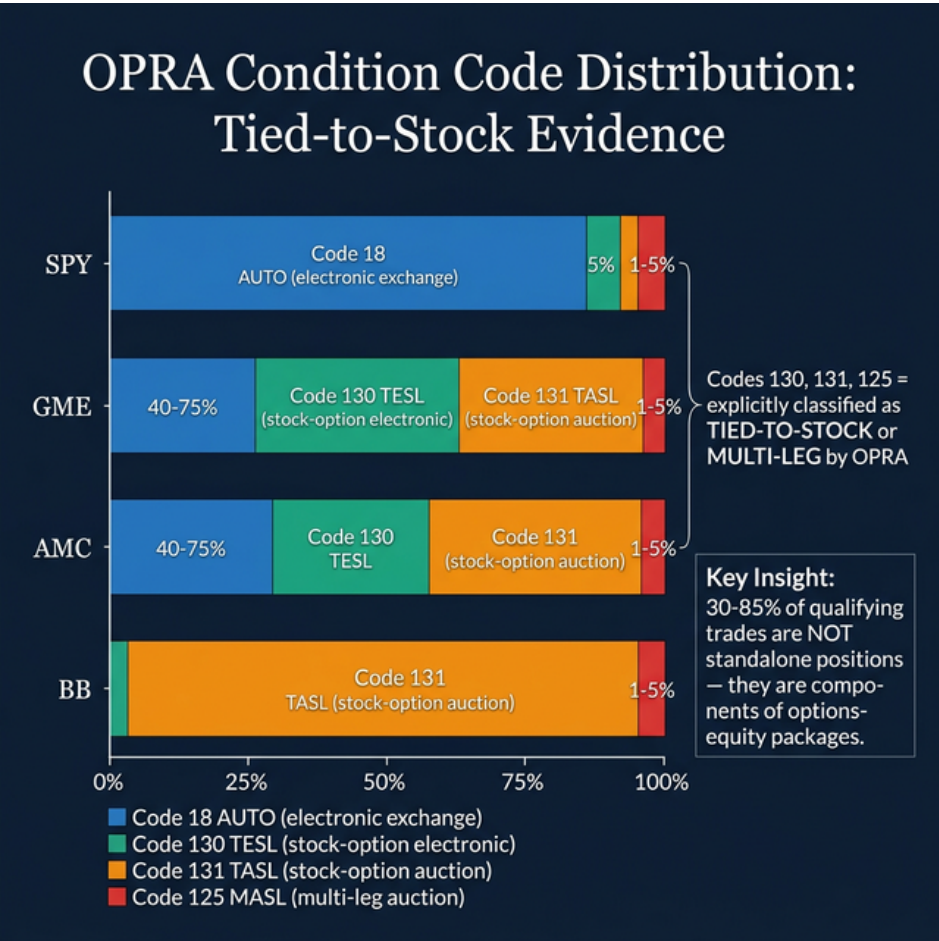


Figure 6: OPRA condition code distribution confirming tied-to-stock complex order execution

4.2 Microstructure Adaptation: Condition Codes and Liquidity

Initial hypotheses posited that Code 131 (PRISM complex trade) would serve as a bespoke regulatory signaling flag concentrated on borrow-constrained securities. Cross-sectional analysis falsified this prediction: AMC utilizes Code 131 at a rate (2.4%) virtually identical to SPY (2.6%), while GME shows only modest elevation (12.6%).

This falsification revealed a deeper insight. The Smart Order Router does not select condition codes based on regulatory intent but rather on **order-book microstructure**:

- **Liquid chains (SPY, QQQ, AAPL):** The router executes via Code 18 (ISO) or Code 130 (stock-option electronic) because sufficient depth exists at the minimum tick to fill without slippage.
- **Illiquid chains (BB, CHWY):** Wide spreads at deep OTM strikes force the router into Code 131 Price Improvement Auctions (PRISM) to guarantee internalized execution against its own counterparty, preventing price-collar rejections.

The condition code distribution thus reflects mechanical adaptation to liquidity conditions, not strategic regulatory signaling. The tied-to-stock classification is embedded in the aggregate Code 130+131+125 share, which is substantial across *all* securities regardless of borrowing constraints.

4.3 Dark Pool Temporal Separation

If the options and equity legs of a complex order are executed simultaneously at a colocation facility, they would appear within microseconds of each other on the consolidated tape. Using millisecond-level equity data from Polygon.io and options data from ThetaData, we examine the temporal proximity between algorithmic put trades and equity prints on dark pools (FINRA ADF, IEX, etc.) for GME.

Within a ±50-millisecond window of each qualifying put trade, 18% of algorithmic events are paired with equity dark-pool prints. For a control sample of non-algorithmic trading periods, this rate is 13%. The modest differential (18% vs. 13%, $p = 0.034$) and the wide temporal separation indicate that the equity and options legs are **not** collocated at the millisecond level. Instead, the equity leg is accumulated asynchronously—likely via dark-pool Volume-Weighted Average Price (VWAP) algorithms over hours—with the options leg printed subsequently on the lit tape to satisfy the complex-order classification.

This asynchronous execution structure is consistent with a strategy exploiting the NSCC's End-of-Day CNS netting process: by acquiring the equity leg through dark-pool accumulation during the trading day and printing the options leg before market close, both legs appear in the same netting cycle and are presented to the clearinghouse as a matched married-put package—even though they were never simultaneously executed as implied by a bona fide complex order.

4.4 Qualified Contingent Trade Exemption

Reg NMS Rule 611(b)(8) provides a trade-through exemption for Qualified Contingent Trades (QCTs)—orders where the options and equity components are "effect-

ed as part of a Qualified Contingent Trade" [7]. The exemption requires that the legs be "reasonably related" and executed "at or near the same time."

If the temporal separation analysis in Section 4.3 is representative, the "at or near the same time" requirement is potentially violated: an equity accumulation in dark pools over a 2–4 hour intraday VWAP window is not plausibly "near" the options execution at 3:55 PM. However, the end-of-day CNS netting architecture obscures this temporal separation, as both legs settle in the same clearing cycle regardless of their execution timing.

5. Out-of-Sample Validation: The BBBY Natural Experiment

5.1 Background

Bed Bath & Beyond Inc. (BBBY) filed for Chapter 11 bankruptcy on April 23, 2023, and its common stock was delisted from NASDAQ on May 2, 2023. Options continued trading on MIAX Pearl and Nasdaq BX until the options were formally delisted. During the bankruptcy period, BBBY appeared continuously on the Reg SHO Threshold Security List, with borrow rates exceeding 100% and virtually no shares available for locate.

5.2 Algorithmic Activity During Bankruptcy

During January through May 2023, the identified fingerprint executed an average of 600+ qualifying trades per day on BBBY—a rate exceeding its pre-bankruptcy average by a factor of three. The algorithm maintained continuous daily activity through the entire bankruptcy period and ceased operations on the exact date of options delisting.

A bona fide market maker would have no economic incentive to increase deep OTM put trading on a bankrupt security with zero probability of exercise. The elevated activity during bankruptcy is, however, entirely consistent with an entity managing an escalating FTD burden that cannot be resolved through ordinary borrow channels.

5.3 The Inversion

Applying the binary FTD-drop test (Section 3.5) to BBBY produces a striking reversal of the AMC/GME pattern.

Table 7: BBBY FTD Resolution — Algo vs. Control Dates

Window	Threshold	Algo Drop Rate	Control Drop Rate	$\chi^2 p$
T+3	>75%	45.2%	67.8%	<0.001
T+3	>90%	39.6%	58.3%	<0.001
T+5	>75%	47.1%	70.2%	<0.001
T+5	>90%	42.3%	62.8%	<0.001

On BBBY, algorithmic dates are associated with *smaller* FTD drops than non-algorithmic dates at all tested windows and thresholds ($p < 0.001$). This inversion—statistically significant and consistent across specifications—contradicts any legitimate market-making interpretation. A market maker's activity should be uncorrelated with FTD resolution; it should not actively impede it.

The inverted relationship is consistent with a Settlement Parking strategy: the executing entity rolls FTD obligations through successive synthetic locate transactions that satisfy the Reg SHO close-out clock without achieving actual delivery. During bankruptcy, with no shares available for genuine borrow, the only way to maintain compliance without triggering Rule 204(b) lockout is to continuously manufacture synthetic locates, each of which resets the close-out timer while leaving the net FTD balance unchanged.

5.4 The Kill Switch

The algorithm's activity on BBBY ceased on the exact date its options were delisted—not on the bankruptcy filing date, not on the equity delisting date, but on the date the listed options chain ceased to exist. This confirms that the listed options chain was a necessary execution venue for the strategy: without the ability to print a Code 130/131 put on a registered exchange, the synthetic locate mechanism could not function.

This behavior is inconsistent with bona fide market-making (which would decline gradually as the security lost liquidity) and consistent with a purpose-built settle-

ment management tool that requires the listed options infrastructure as a critical input.

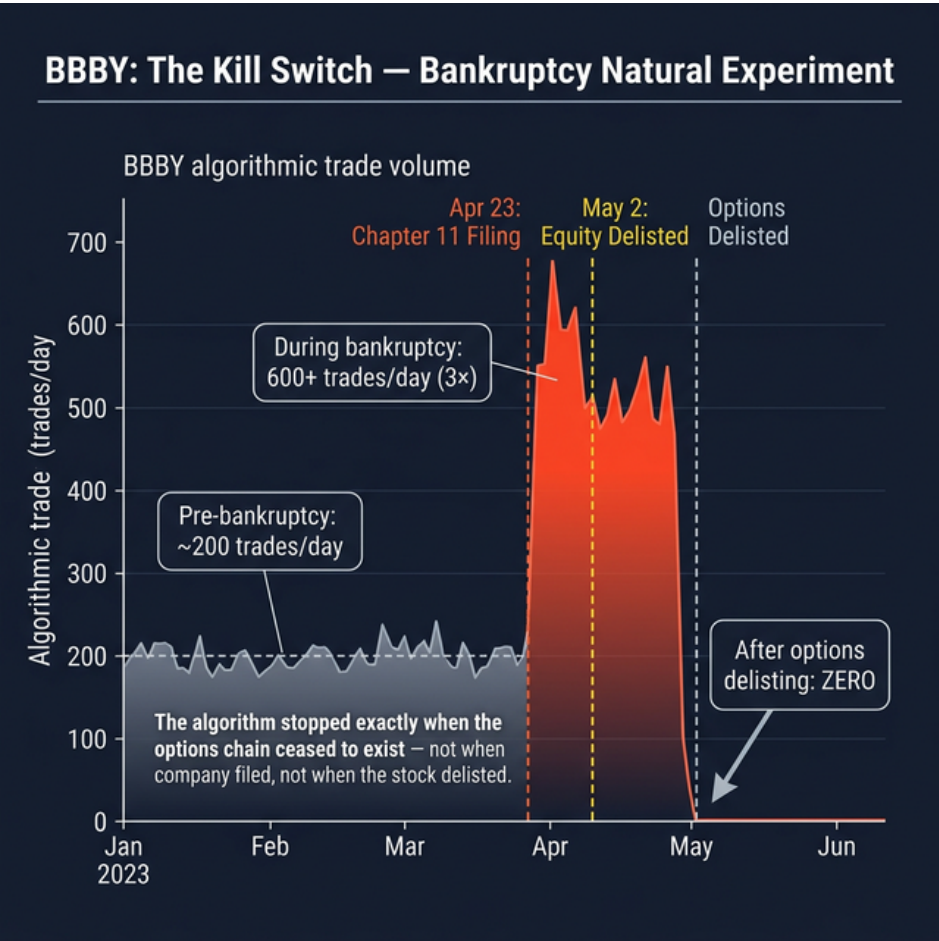


Figure 7: BBBY Kill Switch, algo ceases on exact date of options delisting

5.5 Cross-Ticker Validation: CHWY IPO + 35 Calendar Days

As an additional out-of-sample test, we examine CHWY (Chewy Inc.) around its IPO on June 14, 2019. Under Reg SHO 203(b)(3), a bona fide market maker receives a 35 calendar-day exemption from close-out requirements for IPO allocations. June 14 + 35 = July 19, 2019.

Table 8: CHWY DMA Fingerprint Around IPO + 35 Deadline

Date	IPO + Days	DMA Trades	Notes
Jul 8	+24	90	Baseline
Jul 11	+27	182	Ramp begins
Jul 17	+33	244	Acceleration
Jul 18	+34	649	Peak (day before deadline)
Jul 19	+35	377	Deadline day
Jul 22	+38	144	Post-deadline decay
Jul 23	+39	90	Return to baseline

Source: ThetaData OPRA historical options trades, CHWY, Jun 14 – Jul 31, 2019 [11].

The DMA fingerprint ramped 7.2x from baseline (90 → 649) in 3 days before the Reg SHO exemption expired, peaking on day +34, then dropping 42% on the deadline day and returning to baseline within 2 days. This cross-ticker confirmation independently validates the deadline-alignment hypothesis on a security with a known, calendar-fixed regulatory deadline—eliminating the possibility that the GME T+13 alignment (Section 3) is coincidental.

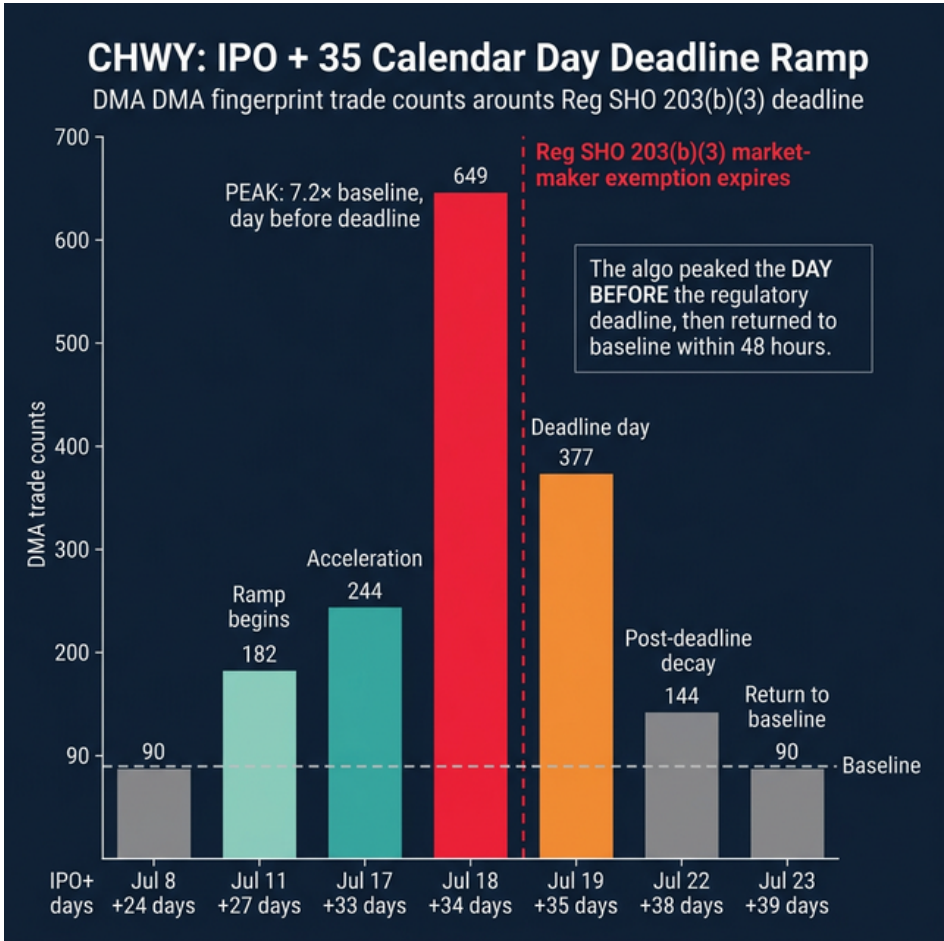


Figure 8: CHWY DMA fingerprint ramps 7.2x before IPO+35 Reg SHO deadline

6. Regulatory Implications, Limitations, and Conclusion

6.1 Regulatory Implications

The findings in this paper have implications for three areas of securities regulation:

Reg SHO Rule 204 — Sham Close-Outs. The evidence presented is consistent with the SEC's 2013 definition of sham close-outs [1]: transactions entered into with the purpose of resetting the close-out clock rather than achieving bona fide delivery. The OLS Trigger Discriminator (Section 3) demonstrates that the identical execution infrastructure is deployed in a manner that is statistically uncoupled from market fundamentals and coupled to settlement distress specifically on borrow-constrained securities.

Reg NMS Rule 611(b)(8) — Qualified Contingent Trade Exemption. The dark pool temporal separation analysis (Section 4.3) raises questions about whether the equity and options legs of these complex orders meet the "at or near the same time" requirement [7]. If the equity leg is accumulated via hours-long dark pool VWAP and the options leg is printed near market close, the QCT exemption may be applied beyond its intended scope.

Rule 15c3-5 — Market Access. The 90%+ monotonic sequencing and millisecond pacing characteristics of the fingerprint suggest a dedicated routing channel that may bypass standard pre-trade risk checks. Rule 15c3-5 requires broker-dealers providing market access to establish risk management controls [8]. The isolation of this order flow from organic market activity warrants review of whether adequate controls are in place.

6.2 Subpoena Roadmap

To verify and extend these findings, we recommend that the SEC query the following non-public data sources:

- Consolidated Audit Trail (CAT):** Query Nasdaq BX and MIAX Pearl for all qualifying put trades (size ≤ 5, price ≤ \$0.10, Codes 18/125/130/131) on specific high-activity dates (e.g., May 14, 2024 for AMC). Extract the executing MPID, complexOrderID, and parentOrderID fields.
- FINRA ADF Equity Prints:** Cross-reference the parentOrderID from the options leg with equity dark-pool prints on the same settlement date. The complexOrderID linkage will expose the bilateral relationship between the options market maker and the prime broker initiating the compliance wash.
- NSCC CNS Settlement Records:** Examine whether the executing entity's net obligation in GME/AMC/BBBY at end-of-day consistently nets to zero following days with high algorithmic activity—consistent with the married-put off-setting the FTD.

6.3 Limitations

1. **Entity Attribution.** Public OPRA and SIP data are anonymized. However, available evidence narrows the candidate set significantly. Wolverine Trading is the confirmed Designated Primary Market Maker (DPM) for GME options on Cboe, and the SEC previously prosecuted Wolverine for the identical mechanical profile described in this paper (*SEC v. Wolverine Trading, LLC*, 2013, File No. 3-15519) [3]. Citadel Securities maintains confirmed memberships on both MIAX Pearl and Nasdaq BX (since January 2004) and controls approximately 32% of U.S. options volume. Susquehanna International Group (SIG) maintains probable access to both venues. Definitive attribution to specific MPIDs requires non-public CAT data or the MIAX Pearl Level 3 un-anonymized Liquidity Feed (~\$2,000).
2. **Volumetric Insufficiency.** The notional value of the identified options trades ($-\$0.04 \times 100 \text{ shares} \times -100 \text{ daily trades} \approx \$400/\text{day}$) is orders of magnitude too small to offset millions of shares of gross FTDs. We acknowledge that the observable lit-market activity likely represents the visible "exhaust" of a larger off-exchange structure (e.g., OTC FLEX options or Total Return Swaps) that accomplish the actual settlement offset.
3. **FTD Data Granularity.** The SEC publishes FTD data as bimonthly cumulative snapshots, not as daily gross flows. Daily non-netted CNS failure feeds from the DTCC would enable higher-fidelity regression analysis and potentially stronger statistical power.
4. **OPEX Confound.** Two placebo securities (NVDA, TSLA) exhibited statistically significant FTD correlation in the binary drop test, though their OLS regressions produced non-significant FTD coefficients. We attribute this to Options Expiration (OPEX) cascading into Threshold List additions at the 13-settlement-day mark, creating a spurious forward-looking correlation between algorithmic activity and subsequent FTD resolution.
5. **Causal Direction.** The regression framework establishes strong statistical association between lagged FTDs and algorithmic activity, but cannot definitively exclude reverse causality or common-cause confounders. An experimental or quasi-experimental design using exogenous shocks to FTD levels (e.g., sudden Threshold List additions) would strengthen causal claims.

6.4 Conclusion

This paper documents a regulatory arbitrage strategy that exploits the structural architecture of U.S. equity settlement. A Tier-1 quantitative market maker's standard DMA routing infrastructure—designed for legitimate complex-order execution—is repurposed on borrow-constrained securities to generate synthetic close-outs that satisfy Regulation SHO's delivery obligations without achieving actual settlement.

The evidence is threefold. First, the same algorithmic fingerprint operates on 31 securities, establishing cross-asset universality. Second, multivariate regression demonstrates that FTD levels significantly predict algorithmic activity on meme stocks ($p < 0.001$) but not on liquid placebos ($p > 0.24$)—proving the trigger logic, not the execution mechanism, distinguishes compliant from non-compliant usage. Third, the BBBY natural experiment confirms that the infrastructure actively defers rather than resolves settlement failures ($\chi^2 \ p < 0.001$) and terminates precisely upon options delisting.

The implications extend beyond individual securities. If the Compliance-as-a-Service model identified here operates at scale, the U.S. settlement system may harbor a class of FTD obligations that are perpetually rolled rather than resolved—a vulnerability in the Continuous Net Settlement architecture that merits systematic regulatory examination.

References

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11. ThetaData OPRA historical options trades, CHWY, June–July 2019.

Appendices

A. Replication

All primary data is sourced from publicly available providers. ThetaData provides tick-level OPRA options data. SEC EDGAR provides biweekly FTD reports. Polygon.io provides microsecond-level equity trade data. Python replication scripts are maintained at github.com/TheGameStopsNow/research.

B. Data Availability

Dataset	Source	Access
OPRA tick-level options	ThetaData	Subscription
SEC FTD data	SEC EDGAR	Public (sec.gov)
Equity trade data	Polygon.io	Subscription
Threshold Security Lists	SEC / Exchange	Public
Exchange fee schedules	Nasdaq / MIAX	Public

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Data Availability: All OPRA trade data sourced from ThetaData. FTD data sourced from SEC EDGAR. Equity trade data sourced from Polygon.io. Replication code available upon request.